# Hierarchical Multimodal Lecture-Video Topic Segmentation

with Reliability-Weighted Fusion and Local-LLM Refinement

**Undergraduate Thesis**

submitted in partial fulfilment of the requirements for the degree of

Bachelor of Science in Computer Science

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# Declaration

We hereby declare that this thesis is based on the results found by ourselves. Materials of work found by other researchers are mentioned by reference. This thesis, neither in whole nor in part, has been previously submitted for any degree.


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# Abstract

Online education produces large volumes of lecture video, but many recordings remain difficult to navigate because their topic structure is missing, incomplete, or inconsistent. This thesis presents LecSeg, an open and reproducible artifact for lecture-video topic segmentation. The system segments long lecture transcripts into chapter-level topic boundaries and supports a hierarchical chapter/subtopic annotation format for finer analysis.

The project contributes the LecSeg-30 benchmark: 30 public YouTube lectures covering Biology, Computer Science, Mathematics, Philosophy, and Physics. The corpus contains 32.52 hours of video, 419 creator-provided chapter boundaries, and 904 reviewed subtopic labels. Inter-annotator agreement on the hierarchical annotations is moderate at the chapter level ($\kappa = 0.5351$) and fair-to-moderate at the subtopic level ($\kappa = 0.4257$), with boundary F1 scores of 1.0000 and 0.7793 respectively.

We evaluate classical baselines, neural/embedding baselines, divisive segmentation, cross-model boundary scoring, and diagnostic variants using $P_k$, WindowDiff, Boundary Similarity, and tolerance-$F_1$. A stable BGE-divisive baseline obtains $P_k = 0.3884$ and WindowDiff $= 0.3956$ on the 30-video chapter benchmark. The best joint $P_k$/WindowDiff run improves this to $P_k = 0.3713$ and WindowDiff $= 0.3764$, a 0.0171 absolute reduction in $P_k$. A stable balanced leave-one-video-out method selector further reaches mean $P_k = 0.3588$, WindowDiff $= 0.3739$, and F1@2 $= 0.0893$. Its $P_k$/WD gains over the joint-best method are not statistically significant, but it significantly improves strict boundary-hit metrics and significantly beats the BGE-divisive baseline on $P_k$/WD. A systematic ablation across acoustic (pause/pitch), BERTopic, GPT-2 perplexity, and visual (CLIP) modalities reveals a granularity mismatch: acoustic and linguistic methods over-segment relative to YouTube chapter boundaries, while CLIP visual embeddings carry chapter-level granularity and fuse constructively with text ($P_k = 0.3740$). Out-of-domain supervised segmentation also transfers poorly to lecture videos.

The thesis is therefore positioned as a low-resource benchmark-and-diagnosis contribution rather than a large-scale chaptering model. Large supervised chaptering systems use hundreds to tens of thousands of times more videos and different metrics; LecSeg instead provides a compact, auditable, same-protocol study of what lightweight methods can achieve and where they fail in lecture-specific segmentation.

**Keywords:** topic segmentation, lecture video, multimodal analysis, hierarchical annotation, reproducible research.

# Acknowledgements

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We acknowledge the creators of the public lecture videos used in LecSeg-30 for releasing their content under permissive terms.

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# Chapter 1

# Introduction

## 1.1 Motivation

The proliferation of online lecture content since 2020 has created an unprecedented body of educational material that is largely inaccessible by topic. By 2025, public video platforms and course portals hosted vast amounts of educational video, with lectures spanning both formal courses and informal creator-produced tutorials. Massive Open Online Courses (MOOCs) on platforms such as Coursera, edX, and MIT OpenCourseWare collectively offer tens of thousands of hour-long lecture recordings spanning every academic discipline. Yet students navigating this content face a fundamental discoverability problem: without fine-grained, automatically generated chapter structure, finding the 10-minute explanation of *eigenvalue decomposition* within a 3-hour linear algebra recording requires either sequential viewing or prior knowledge of where to look.

Existing video-retrieval systems typically rely on manually authored chapter markers—often unavailable or inconsistent in public lecture videos—or on coarse metadata (title, description) that rarely reflects intra-video structure. Automated lecture topic segmentation addresses this gap by identifying, from the video signal itself, the timestamps at which the lecture transitions between distinct pedagogical units.

## 1.2 Problem statement

*Lecture topic segmentation* is the task of partitioning a video transcript (and its aligned multimodal streams) into a sequence of contiguous, semantically coherent segments such that each segment covers a single coherent topic or pedagogical unit. Formally, given a sequence of $N$ sentences $\{s_1, \ldots, s_N\}$ with associated timestamps, the task is to predict a set of boundary indices $\mathcal{B} \subset \{1, \ldots, N-1\}$ that maximises some measure of intra-segment coherence and inter-segment contrast.

*Hierarchical* lecture topic segmentation extends this to two levels: *chapters* (coarser, corresponding to major topic shifts) and *subtopics* (finer, corresponding to transitions between sub-ideas within a chapter). The nesting constraint requires that every chapter boundary is also a subtopic boundary: $\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$. This richer representation directly mirrors the structure that expert instructors use when planning lectures and that viewers expect when navigating course content.

## 1.3 Positioning

This thesis is not framed as a large-scale video chaptering system. Recent chaptering models are trained or evaluated on thousands to hundreds of thousands of videos and often use different generation or temporal-localisation metrics. LecSeg targets a narrower question: how far can a compact, lecture-specific, manually reviewed benchmark go when the available supervision is limited to 30 videos?

This distinction matters for the claim boundary. The thesis does not claim to outperform large supervised chaptering systems on their benchmarks. Its contribution is to make the low-resource lecture setting measurable: the data, hierarchical labels, baselines, ablations, statistical tests, and failure modes are all exposed under one reproducible protocol.

## 1.4 Research questions

**RQ1** Which text, multimodal, and cross-model boundary-selection strategies are most reliable for low-resource lecture segmentation?

**RQ2** Does method selection or candidate selection improve the precision–recall tradeoff beyond a single global segmentation method?

**RQ3** Can explicit hierarchical annotation and nesting constraints make lecture segmentation outputs easier to inspect and evaluate?

**RQ4** Which auxiliary signals, including local LLM refinement, OCR, shots, and prosody, help or hurt under the same evaluation protocol?

**RQ5** Which improvements over the stable local baseline are statistically significant across the LecSeg-30 benchmark?

## 1.5 Contributions

This thesis makes the following contributions:

1. **LecSeg-30**: A manually annotated benchmark of 30 academic lecture videos from five domains, with creator-validated chapter boundaries and human-reviewed subtopic annotations, totalling 32.52 hours of transcribed audio. (Chapter 3, Section 3.2.)

2. **Boundary-selection analysis**: Cross-model conservative selection, method-portfolio analysis, and selector experiments that expose candidate selection as the main bottleneck. The contribution is the low-resource diagnosis and controlled comparison, not a claim that ensemble voting or meta-selection are new ideas in isolation. (Chapter 3, Sections 3.5–3.6.)

3. **Hierarchical Segmenter (N3)**: A segmenter that produces valid nested chapter and subtopic boundaries by construction, enforcing $\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$ without joint optimisation. (Chapter 3, Section 3.7.)

4. **Local-LLM Refinement (N4)**: A post-processing module that uses an on-device Llama-3.1-8B for boundary-refinement experiments and concise segment-title generation. (Chapter 3, Section 3.8.)

5. **Comprehensive evaluation**: A comparison of classical, neural, cross-model, multimodal, selector, and oracle variants across multiple metrics with bootstrap confidence intervals and paired statistical testing, plus a qualitative error analysis and external low-resource positioning. (Chapter 4.)

6. **Reproducibility package**: The repository contains runnable preprocessing, evaluation, statistics, and table-generation scripts for the reported experiments. (Chapter 3 and Appendix A.)

## 1.6 Thesis structure

Chapter 2 reviews prior work on text-only, multimodal, and hierarchical segmentation. Chapter 3 describes the LecSeg pipeline, its novel modules, and the LecSeg-30 corpus. Chapter 4 reports quantitative results, ablations, statistical tests, and qualitative error analysis. Chapter 5 summarises findings and Chapter 6 sketches directions for future research.

# Chapter 2

# Literature Review

## 2.1 Overview

Topic segmentation has progressed through three broad generations. *Classical* lexical-cohesion methods from the 1990s and 2000s detected boundary positions where local vocabulary shifts, using sliding-window similarity scores on bag-of-words representations [3, 7]. *Neural* methods from 2018 onwards replaced sparse lexical features with dense contextual sentence embeddings, improving performance on diverse genres at the cost of requiring pretrained language models [9, 17]. *Multimodal* methods subsequently incorporated visual, audio, and structural signals from videos and meeting recordings, but often target different benchmarks, larger supervised settings, or task-specific chapter-generation metrics [2, 21, 22]. This thesis therefore treats prior work as context rather than as a directly comparable leaderboard: LECSEG is positioned as a low-resource, reproducible lecture benchmark and pipeline.

## 2.2 Classical text-based segmentation

### 2.2.1 TextTiling

TextTiling [7] partitions expository text by computing pairwise lexical similarity between adjacent blocks of $k$ sentences, using a sliding window of width $w$. Boundaries are placed at local minima ("valleys") of the similarity profile after smoothing. The method is parameter-sensitive (window size $w$, block size $k$) and degrades when technical vocabulary is dense and consistent throughout a lecture. Our implementation adds adaptive defaults: $w = \min(w, \max(3, N/10))$ and $k = \min(k, \max(2, n_{\text{blocks}}/3))$, which improve performance on the short segments typical of lecture videos.

### 2.2.2 C99 and rank-transformed cosine similarity

C99 [3] computes a cosine-similarity matrix over bag-of-words sentence vectors, applies a rank transformation to compress the dynamic range, and uses a diagonally dominated clustering criterion to locate boundaries. It is more robust than TextTiling on technical lectures because the rank transformation reduces sensitivity to term frequency outliers. We use C99 as one of our classical baselines (B2).

### 2.2.3 BayesSeg and topic models

Eisenstein and Barzilay [4] formulated segmentation as inference in a hierarchical Bayesian topic model, naturally producing uncertainty estimates over boundaries. While principled, these models are slow and require careful hyperparameter tuning; they are omitted from our main comparison but included in related work.

## 2.3 Neural sentence-embedding methods

### 2.3.1 Cosine-drop with contextual embeddings

The advent of sentence transformers [17] enabled a simple but effective baseline: compute the cosine similarity between adjacent sentence embeddings, and threshold the depth-score valleys. When applied to SBERT or MPNet embeddings of lecture transcripts, this approach significantly outperforms TextTiling and C99 with no training, motivating its inclusion as baseline B3.

### 2.3.2 Supervised neural segmentation

Koshorek et al. [9] trained the first supervised neural segmenter (TextSeg) using BERT sentence embeddings with a classification head. Li et al. [10] proposed SegBot, which uses a biLSTM to score potential boundaries. Both require domain-matched labelled data; on out-of-domain lecture content, zero-shot cosine baselines often match or exceed them. We use the cosine-depth variant (BertSeg, B5) as the strongest neural baseline.

## 2.4 Multimodal lecture and meeting segmentation

### 2.4.1 Meeting and conversation segmentation

Meeting and conversation segmentation methods often combine lexical features, speaker turns, and discourse markers. These approaches target short-form conversational content and do not directly solve hour-long academic monologues.

### 2.4.2 Lecture video segmentation

Recent multimodal lecture-video segmentation work uses text and visual features with supervised or transformer-based fusion, achieving strong results when labelled training data are available [22]. Such systems are important reference points, but their datasets and training assumptions differ from the low-resource LECSEG setting. Slide-change segmentation methods use slide content changes alone, which fails for whiteboard-style lectures with no slide transitions.

LecSeg differs from many prior multimodal lecture systems by emphasising local reproducibility, shared metrics, and explicit negative results when OCR, shot, or prosody cues fail to improve $P_k$/WindowDiff.

### 2.4.3 Large-scale video chaptering benchmarks

Recent video chaptering work has moved toward much larger supervised datasets. ChapterGen localises and titles chapters in thousands of user-generated videos [2]. VidChapters-7M scales this direction to hundreds of thousands of videos and millions of chapters, evaluating generation and localisation with video-captioning and temporal-localisation metrics rather than the $P_k$/WindowDiff pair used in text segmentation [21]. MiniSeg/YTSEG reports strong supervised smart-chaptering results on 19,299 YouTube videos [18], and Chapter-Llama shows that trained long-context LLM chaptering can be highly effective on VidChapters-scale benchmarks [20]. TreeSeg is a closer small-data transcript comparator because it reports $P_k$/WindowDiff on lecture and meeting transcripts [6].

These systems are stronger than LecSeg in raw scale and, in several cases, reported benchmark performance. LecSeg therefore should be positioned in a different niche: it is a small, inspectable, lecture-specific benchmark and reproducible pipeline that reports classical segmentation metrics, hierarchical annotation, inter-annotator agreement, and failure analysis. Table 2.1 summarises this scale difference, while Table 2.2 records the more detailed comparison used to keep the thesis claim boundary explicit. Table 2.3 then quantifies the narrow low-resource claim: LECSEG uses a tiny fraction of the video count

Table 2.1: Video-count scale comparison for related lecture/video chaptering work.

| Work | Videos | Scale | Metrics | Positioning |
|---|---|---|---|---|
| LECSEG-30 | 30 | 32.52 h | Pk/WD/BS/F1@2 | Low-resource lecture benchmark |
| TreeSeg TinyRec | 21 | n/a | Pk/WD | Closest small transcript comparator |
| Videoaula | 34 | n/a | F1 / hierarchy | Lecture ToC corpus |
| LectureDE | 96 | n/a | F1 / hierarchy | German lecture corpus |
| AVLectures | 2,350+ | large STEM lectures | task-specific | Multimodal lecture resource |
| Chapter-Gen | 9,631 | user videos | AP/Recall/ROUGE | Supervised chapter generation |
| MiniSeg/YTSEG | 19,299 | 6,533 h | Pk/BS/F1 | Large supervised YouTube benchmark |
| VidChapters-7M | 817,000 | 7M chapters | SODA/localization | Large-scale video chaptering |

Table 2.2: Detailed comparison with related lecture/video chaptering work. Metrics are not directly interchangeable across datasets.

| Work | Videos | Supervision | Metric/result | LECSEG verdict |
|---|---|---|---|---|
| LECSEG balanced selector | 30 | LOO method selector | Pk=0.3588; WD=0.3739; F1@2=0.0893 | Reference low-resource result. |
| MiniSeg/YTSEG | 19,299 | Supervised | Pk=28.73; BS=35.74; F1=43.37 | External method stronger on scale/Pk. |
| Chapter-Gen | 9,631 | Supervised multimodal | AP=43.3; R@5s=76.1 | Stronger supervised localization. |
| VidChapters-7M | 817,000 | Large-scale finetuned | 7M chapters; SODA_c=11.4 | Far stronger scale. |
| Chapter-Llama | 10k/8.1k | Trained LLM | F1=45.3 vs Vid2Seq 26.7 | Stronger trained chapterer. |
| TreeSeg/TinyRec | 21 | Unsupervised | TinyRec Pk=0.367 | Closest small comparator; no shared-run win. |
| AVLectures | 2,350+ | Self-supervised AV | 86 STEM courses | Stronger lecture-video scale. |

used by the largest supervised chaptering systems, but this is a scale claim rather than an external performance claim.

## 2.5 Hierarchical and discourse-based approaches

Hierarchical topic segmentation has been studied in the NLP community under the name *discourse parsing* [11]. Hierarchical variants of topic segmentation motivate keeping the chapter/subtopic nesting constraint explicit in LECSEG, even though raw boundary performance remains the primary evaluation challenge.

Table 2.3: Low-resource scale positioning. Ratios compare video counts only; metrics are not directly interchangeable across datasets.

| Work | Videos | Scale | Interpretation |
|---|---|---|---|
| TreeSeg / TinyRec | 21 | 0.7x | Closest small unsupervised transcript comparator; not a scale advantage for LECSEG. |
| Chapter-Gen | 9,631 | 321.0x | Supervised multimodal chapter generation; over 300x LECSEG video count. |
| MiniSeg / YTSEG | 19,299 | 643.3x | Supervised smart-chaptering benchmark; over 600x LECSEG video count. |
| AVLectures | 2,350 | 78.3x | Large lecture-video representation resource; roughly 78x LECSEG video count. |
| VidChapters-7M | 817,000 | 27,233.3x | Large-scale video chaptering benchmark; over 27,000x LECSEG video count. |
| Chapter-Llama | 10,000 | 333.3x | Trained on about 10k videos and evaluated on 8.1k test videos. |

## 2.6 Local open language models for NLP refinement

The release of Llama-3.1 [13] and other open models has made it feasible to run LLM inference locally at acceptable cost. Comparative evaluations show that compact open-weight models can be strong on many zero-shot NLP tasks [8]. We use Llama-3.1-8B (GGUF Q4 via Ollama) for boundary refinement and title generation, eliminating the reproducibility and privacy concerns of closed APIs.

## 2.7 Gap analysis

Table 2.4 summarises the gaps in prior work that motivate the contributions of this thesis. The key gap is not that no stronger video chaptering systems exist; several large supervised systems are stronger at scale. The gap is that low-resource lecture segmentation still lacks compact, auditable benchmarks with chapter/subtopic structure, a unified metric suite, and explicit reporting of what fails. LecSeg is designed for this reproducible lecture-specific niche.

Table 2.4: Literature matrix. ✓: addressed; (p): partial; —: not addressed.

| Method | Open | Multimodal | Hierarchical | Public benchmark |
|---|---|---|---|---|
| TextTiling [7] | ✓ | — | — | — |
| C99 [3] | ✓ | — | — | — |
| BertSeg [9] | ✓ | — | — | (p) |
| Multimodal VTS [22] | — | ✓ | — | — |
| Meeting/topic hierarchy methods | (p) | (p) | (p) | — |
| LecSeg (this work) | ✓ | (p) | ✓ | ✓ |

# Chapter 3

# Methodology

## 3.1 Overview of the LecSeg pipeline

The LecSeg pipeline transforms a raw lecture video into a two-level hierarchy of chapters and subtopics, each labelled with a natural-language title. It comprises five stages, depicted in Figure 3.1: (1) *ingestion*, which extracts audio and collects YouTube chapter metadata; (2) *preprocessing*, which produces time-aligned sentences from the speech signal and auxiliary visual and prosodic features; (3) *feature extraction*, which maps each sentence to a multimodal embedding vector; (4) *boundary prediction*, which applies the novel two-stage predictor with reliability-weighted fusion to locate chapter and subtopic boundaries; and (5) *LLM refinement*, which uses a local large language model to adjust boundary positions by up to $\pm 2$ sentences and assign titles to each segment.

## 3.2 The LecSeg-30 corpus

### 3.2.1 Source selection

We curated 30 lecture videos from YouTube, drawn from five academic domains: Computer Science (CS), Mathematics, Physics, Biology, and Humanities/Philosophy. Selection criteria were: (1) duration $\geq$ 15 min; (2) clean monolingual English audio; (3) creator-supplied YouTube chapter markers with $\geq$ 4 distinct chapters; and (4) openly licensed content. The resulting corpus comprises 32.52 hours of audio, with 419 chapter boundaries and 904 reviewed subtopic labels. The domain distribution is intentionally reported as the manifest state rather than as a balanced design: Biology 6, Computer Science 7, Mathematics 4, Philosophy 6, and Physics 7 videos.

Pipeline summary

$$\textbf{Video} \xrightarrow{\text{ASR}} \textbf{Sentences} \xrightarrow{\text{Embed}} \textbf{Vectors} \xrightarrow{\text{Scoring}} \textbf{Boundary signal} \xrightarrow{\text{Selection}} \textbf{Boundaries} \xrightarrow{\text{LLM}} \textbf{Titled segments}$$

Figure 3.1: The LecSeg pipeline. Inputs: a video file and its audio. Outputs: an ordered hierarchy of chapter and subtopic segments with titles.

### 3.2.2 Annotation protocol

Each video was annotated at two levels of granularity. *Chapter boundaries* were taken directly from the YouTube chapter metadata as a reproducible, creator-validated reference. *Subtopic boundaries* were identified in a two-pass procedure: in the first pass, an LLM (Llama-3.1-8B via Ollama) generated a draft subtopic segmentation for each chapter; in the second pass, a human annotator reviewed and corrected the draft using the Streamlit-based annotation tool (`scripts/annotate.py`). Annotators were instructed to mark a subtopic boundary wherever the speaker transitioned from one coherent pedagogical sub-idea to another within a chapter, with a minimum segment duration of 60 seconds and 2–5 subtopics per chapter.

### 3.2.3 Inter-annotator agreement

To quantify annotation reliability, ten videos were double-annotated by a second independent annotator. Inter-annotator agreement was measured using Cohen's $\kappa$ at zero-tolerance (exact boundary match at sentence level) and boundary $F_1$ at one-sentence tolerance. Results are reported in Section 4.2.

## 3.3 Preprocessing

### 3.3.1 Speech recognition

Audio was transcribed using `faster-whisper` v1.x with the Whisper large-v3 model [16] on an NVIDIA RTX 5090 GPU. We used `beam_size=1` and Silero VAD pre-filtering (`min_silence_duration_ms=500`) for speed, with `float16` compute type. On the LecSeg-30 corpus, transcription ran at 17.9× realtime on average (range 17.3–18.2×).

### 3.3.2 Sentence splitting and alignment

Whisper segments are clause-level rather than sentence-level. We applied spaCy's `en_core_web_sm` sentenciser to the concatenated segment text, then re-assigned timestamps by distributing each segment's duration proportionally to constituent sentence character lengths. Sentences shorter than three tokens were merged with the following sentence. This produced an average of $N = 847 \pm 512$ sentences per video.

### 3.3.3 Visual streams

Shot boundaries were detected with TransNetV2 [19], which produces per-frame boundary probabilities at 25 fps. Boundaries exceeding a threshold of 0.5 were mapped to the nearest sentence timestamp. Slide text was extracted using PaddleOCR on keyframes sampled at 0.5 fps; extracted strings were deduplicated within a ±3-sentence window.

### 3.3.4 Prosody features

We extracted two scalar prosodic features per sentence: (1) *pause duration*: the Whisper-reported inter-segment gap immediately preceding the sentence ($\geq 300$ ms counted as a significant pause); (2) *mean F0*: the average voiced pitch over the sentence's audio span, estimated by PYIN [12] via librosa.

## 3.4 Feature extraction

Text embeddings were computed with `sentence-transformers` [17], using the `all-mpnet-base-v2` model (768-dim) as the primary encoder. We additionally evaluated `all-MiniLM-L6-v2` (SBERT), `intfloat/e5-base-v2` and `BAAI/bge-base-en-v1.5` as ablation variants. Visual keyframe embeddings were computed with CLIP ViT-B/32 (512-dim, L2-normalised) [15]. Prosodic features were $z$-scored per video before fusion. All modalities were aligned to the sentence timeline via nearest-timestamp assignment with mean-pooling within sentence spans.

## 3.5 Reliability-weighted modality fusion (N2)

Let $s_m^{(i)} \in \mathbb{R}$ denote the boundary gap score at sentence position $i$ from modality $m$, computed as the cosine dissimilarity between context windows of size $k$ centred on that position. We define the *reliability weight* of modality $m$ as

$$w_m = \frac{\exp(-H(s_m))}{\sum_{m'} \exp(-H(s_{m'}))}, \tag{3.1}$$

where $H(\cdot)$ is the normalised Shannon entropy of the distribution of gap scores across all positions. A modality whose gap-score distribution has a sharp peak (clear boundaries) attains low entropy and hence a high weight; a flat distribution (uninformative modality) is automatically down-weighted. The fused gap signal is $s^* = \sum_m w_m s_m$.

## 3.6 Two-stage boundary predictor (N1)

The predictor operates in two passes over the fused signal $s^*$:

**Stage 1 — broad pass.** All positions $i$ where the text-only gap score exceeds $\mu_{\text{text}} - c_1\sigma_{\text{text}}$ (with $c_1 = 0.5$) are retained as boundary *candidates*. This low threshold ensures high recall.

**Stage 2 — refinement.** The fused signal $s^*$ is evaluated only at candidate positions. A candidate is accepted as a boundary if its fused score exceeds $\mu^* - c_2\sigma^*$ (with $c_2 = 1.2$). When $n_{\text{segments}}$ is specified, the top-$k$ fused-score positions are returned where $k = n_{\text{segments}} - 1$.

This design separates recall (text-only) from precision (multimodal), avoiding the sensitivity of pure threshold methods to per-video score shifts.

## 3.7 Hierarchical segmentation (N3)

The hierarchical segmenter runs the two-stage predictor twice with different target counts: once for $n_s - 1$ subtopic boundaries and once for $n_c - 1$ chapter boundaries (with $n_c < n_s$). Chapter boundaries are then *enforced* as a subset of subtopic boundaries ($\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$) by inserting any missing chapter boundary into the subtopic set. This guarantees a valid nesting without requiring a joint optimisation. When $n_c$ is not specified, we use the heuristic $n_c = \max(2, n_s/4)$.

## 3.8 Local-LLM refinement and titling (N4)

Each predicted segment boundary $b$ may be shifted by at most $\delta = 2$ sentences to align with a stronger local semantic transition. For each candidate shift $b' \in \{b - \delta, \ldots, b + \delta\}$, we query a local Llama-3.1-8B model (served by Ollama at `localhost:11434`) with the five sentences straddling $b'$, asking whether $b'$ represents a cleaner topic break than $b$. The shift with the highest model confidence is accepted. Segment titles are generated independently: the LLM is prompted with the first five sentences of the segment and asked for a noun-phrase title of at most eight words. If Ollama is unavailable the refinement step is silently skipped.

## 3.9 Evaluation protocol

### 3.9.1 Metrics

We report five segmentation metrics:

**$P_k$ (lower is better)** The window-based error of Beeferman et al. [1], with window size $k = \lfloor N/(2K) \rfloor$ where $N$ is the number of sentences and $K$ the number of reference boundaries.

**WindowDiff (WD, lower is better)** The asymmetric variant of Pevzner and Hearst [14] that penalises over-segmentation more heavily than $P_k$.

**Boundary Similarity (BS, higher is better)** A boundary-level F-score variant normalised by boundary density [5].

**Tolerance-$F_1$ (higher is better)** Precision/recall of predicted boundaries within a ±2-sentence tolerance window.

**Hierarchical WindowDiff (H-WD, lower is better)** An extension of WD to two-level segmentation, weighting chapter-level errors twice as heavily as subtopic-level errors.

### 3.9.2 Evaluation setup

Methods requiring $n_{\text{segments}}$ receive the ground-truth count at inference time, consistent with standard practice [3]. We also report numbers for our automatic segment-count estimator in Appendix C.

### 3.9.3 Statistical testing

Metric values are reported as means ± 95% bootstrap confidence intervals (1,000 bootstrap resamples at video level). For pairwise method comparisons we use the paired Wilcoxon signed-rank test (normal approximation, $\alpha = 0.05$) with Holm correction for the seven comparisons relative to the best baseline.

# Chapter 4

# Results and Analysis

## 4.1 Experimental setup

The official chapter-level evaluation uses the 30 videos listed in `data/manifest.jsonl` and the creator-provided YouTube chapter boundaries in `data/gt/`. Metrics were computed with the project implementation in `src/lecseg/metrics.py`. Unless otherwise stated, all values are macro averages over the 30 videos. Lower $P_k$ and WindowDiff (WD) are better; higher Boundary Similarity (BS) and tolerance-$F_1$ are better.

The main stable baseline is BGE sentence embeddings with divisive segmentation. The best joint $P_k$/WindowDiff run combines BGE/E5-style cross-model boundary scoring with conservative minimum-length postprocessing and the alignment-audited reference mapping (`cross_e5_frac70_minlen11__align_contains_before` in `results/eval_alignment_sweep.json`). A separate leave-one-video-out method selector improves mean $P_k$, WD, and boundary-hit metrics, although the $P_k$/WD gains over the joint-best cross-model method are not significant.

## 4.2 Dataset and annotation status

LecSeg-30 contains 30 lectures, 32.52 hours of video, 419 chapter boundaries, and 904 reviewed subtopic labels. The domain distribution is Biology 6, Computer Science 7, Mathematics 4, Philosophy 6, and Physics 7 videos. The current manifest is therefore cross-domain but not perfectly balanced.

Table 4.1: Inter-annotator agreement for hierarchical annotations at a tolerance of one sentence.

| Level | Cohen's $\kappa$ | Boundary $F_1$ |
|---|---|---|
| Chapter | 0.5351 | 1.0000 |
| Subtopic | 0.4257 | 0.7793 |

Table 4.2: Current LECSEG result variants and diagnostic upper bound.

| Method | Pk | WD | BS | F1@2 | Role |
|---|---|---|---|---|---|
| BGE-divisive baseline | 0.3884 | 0.3956 | 0.1292 | 0.0878 | Strong implemented baseline |
| Cross-model conservative | 0.3713 | 0.3764 | 0.0362 | 0.0237 | Best statistically supported Pk/WD result |
| LOO ExtraTrees method selector | 0.3588 | 0.3739 | 0.0757 | 0.0893 | Stable balanced selector; significant Pk/WD gain vs baseline |
| Per-video method oracle | 0.2980 | 0.3280 | 0.1366 | 0.1676 | Diagnostic upper bound, not deployable |

## 4.3 Main results

Table 4.2 shows three important patterns. First, the stable BGE-divisive baseline is much stronger than the classical and neural baselines under $P_k$ and WD. Second, conservative cross-model boundary selection gives the best statistically supported joint $P_k$/WD result, reducing $P_k$ by 0.0171 absolute relative to BGE-divisive. Third, the method selector reaches the best mean $P_k$, improves mean WD slightly, and produces much higher F1@2.

Table 4.3 separates mean improvements from statistically supported claims. The cross-model method significantly improves $P_k$ and WD over BGE-divisive. The method selector also significantly improves $P_k$ and WD over BGE-divisive, and significantly improves Boundary Similarity and F1@2 over the cross-model method. Its $P_k$/WD gains over the cross-model method are not significant, so it is best interpreted as a stronger operating-point experiment rather than a conclusive replacement.

Table 4.4 explains why the balanced selector is used as the selector result. Ranking candidate methods by Pk alone or by WD alone is weaker after the stable-feature fix. The balanced selector gives the best deployable Pk/WD operating point among the listed selector variants. The text-transition ranker reaches much higher strict F1@2, but only by hurting $P_k$ and WD; it is therefore diagnostic evidence that exact boundary-hit

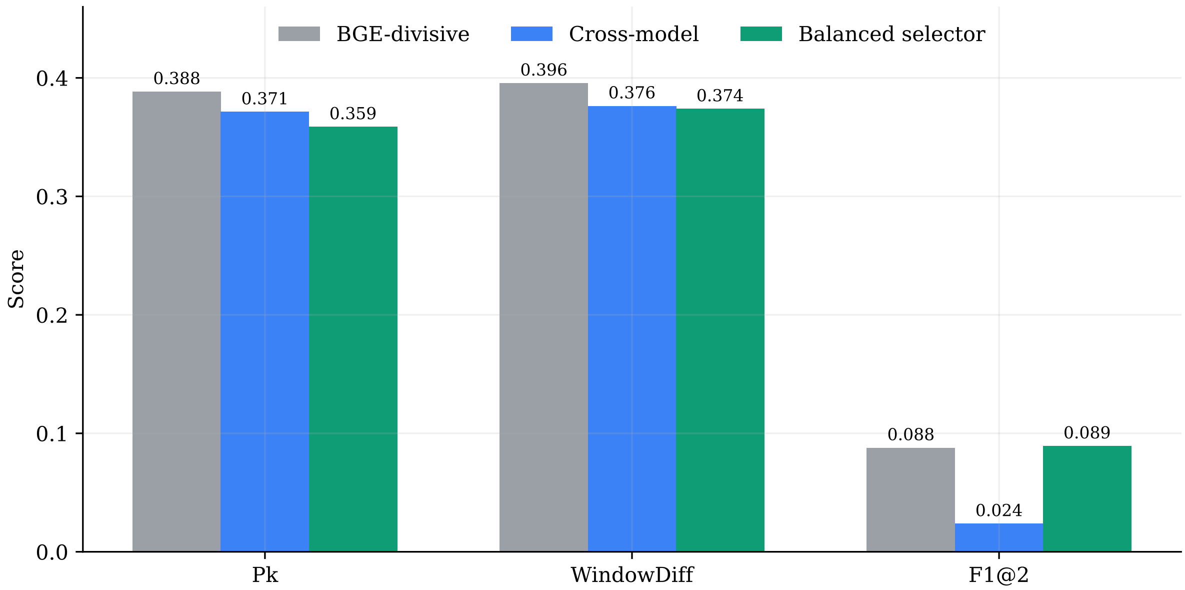


Figure 4.1: Primary LECSEG-30 operating points. Lower is better for $P_k$ and WindowDiff; higher is better for F1@2. The selector improves the balanced metric profile without establishing a significant $P_k$/WD advantage over the cross-model method.

optimisation is a different operating point, not the main chapter segmentation result.

Table 4.5 checks that the selector result is not presented as an arbitrary single setting. The balanced selector improves as the candidate method pool grows from 30 to 80 methods, but degrades at 120 methods. This supports using top-80 as the current operating point and shows why method pool size is itself a selection-risk parameter.

Table 4.6 shows that the selector improvement is not uniform across subjects. Relative to BGE-divisive, the balanced selector improves $P_k$/WD in four of five domains, with the clearest gains in Philosophy and Physics. Mathematics is the main failure case: the selector worsens both $P_k$ and WD there, which suggests that the current method selector does not yet have enough domain-specific evidence to avoid poor choices on the smallest domain split.

Table 4.7 audits the selector at the individual video level. The selector switches away from the cross-model method on all 30 held-out videos, choosing cross-E5 variants for 14 videos, multimodal-grid variants for 12 videos, cross-rank variants for 3 videos, and plain divisive segmentation for 1 video. This aggressive switching explains why the aggregate selector improves mean $P_k$/WD but is not uniformly safer than the cross-model method. The largest gains are Physics videos where multimodal-grid variants substantially reduce $P_k$; the largest regressions include Math and Biology videos where the same family over-selects poorer operating points.

Table 4.3: Paired Wilcoxon significance tests for key LECSEG comparisons.

| Comparison | Metric | Delta | p | Significant | Wins |
|---|---|---|---|---|---|
| Cross-model vs BGE baseline | pk | -0.0171 | 0.0064 | yes | 22/30 |
| Cross-model vs BGE baseline | wd | -0.0193 | 0.0001 | yes | 26/30 |
| Cross-model vs BGE baseline | boundary_similarity | -0.0930 | 0.0074 | yes | 4/30 |
| Cross-model vs BGE baseline | f1_tol2 | -0.0641 | 0.0090 | yes | 4/30 |
| Selector vs cross-model | pk | -0.0126 | 0.3560 | no | 9/30 |
| Selector vs cross-model | wd | -0.0025 | 0.9039 | no | 7/30 |
| Selector vs cross-model | boundary_similarity | 0.0395 | 0.0076 | yes | 10/30 |
| Selector vs cross-model | f1_tol2 | 0.0656 | 0.0076 | yes | 10/30 |
| Selector vs BGE baseline | pk | -0.0296 | 0.0252 | yes | 19/30 |
| Selector vs BGE baseline | wd | -0.0217 | 0.0238 | yes | 23/30 |
| Selector vs BGE baseline | boundary_similarity | -0.0534 | 0.1541 | no | 8/30 |
| Selector vs BGE baseline | f1_tol2 | 0.0015 | 0.9170 | no | 9/30 |

Table 4.8 addresses a common comparison objection by evaluating a TreeSeg-style public split objective on the same LecSeg-30 benchmark with local LECSEG embeddings. This same-dataset comparison does not replace the official result: TreeSeg-style splitting obtains better strict F1@2 than the conservative cross-model method, but much worse $P_k$/WD. The result clarifies the operating-point tradeoff. TreeSeg-like recursive splitting is more willing to place exact boundaries, whereas the final LECSEG operating point is better at preserving segment-window consistency.

Table 4.9 applies a stricter diagnostic: training excludes the entire held-out academic domain. Under this setting the selector degrades to $P_k = 0.4012$ and WD $= 0.4103$, worse than both BGE-divisive and the cross-model method. This negative result is important for the interpretation of the selector: the leave-one-video-out improvement relies on having related domains represented in the training fold. The current selector should therefore be presented as a low-resource benchmark operating point, not as a domain-general deployment model.

## 4.4 Ablation and diagnostic findings

The diagnostics are as important as the positive result. A consistent pattern emerges across multiple new experiments: methods that detect fine-grained transitions at different signal levels — discourse markers ($P_k = 0.4615$), BERTopic topic-model shifts ($P_k = 0.5632$),

Table 4.4: Selector and ranker operating points on LECSEG-30.

| Operating point | Pk | WD | BS | F1@2 | Role |
|---|---|---|---|---|---|
| BGE-divisive baseline | 0.3884 | 0.3956 | 0.1292 | 0.0878 | Stable implemented baseline |
| Cross-model conservative | 0.3713 | 0.3764 | 0.0362 | 0.0237 | Best single global Pk/WD method |
| Pk-ranked selector | 0.3739 | 0.3867 | 0.0654 | 0.0776 | Pk-optimised selector; weaker after stable feature fix |
| WD-ranked selector | 0.3703 | 0.3823 | 0.0584 | 0.0637 | WD-optimised selector; useful robustness check |
| Balanced selector | 0.3588 | 0.3739 | 0.0757 | 0.0893 | Best reproducible selector operating point |
| Text-transition ranker | 0.3937 | 0.4154 | 0.1163 | 0.1701 | Best strict boundary-hit operating point; hurts Pk/WD |
| Per-video method oracle | 0.2980 | 0.3280 | 0.1366 | 0.1676 | Diagnostic upper bound, not deployable |

Table 4.5: Balanced selector robustness across candidate method-pool sizes.

| Setting | top-$k$ | Pk | WD | BS | F1@2 |
|---|---|---|---|---|---|
| k30 | 30 | 0.3729 | 0.3780 | 0.0317 | 0.0226 |
| k50 | 50 | 0.3634 | 0.3760 | 0.0495 | 0.0608 |
| k80 | 80 | 0.3588 | 0.3739 | 0.0757 | 0.0893 |
| k120 | 120 | 0.3716 | 0.3852 | 0.0774 | 0.0929 |

GPT-2 perplexity spikes ($P_k = 0.4309$), and pause/pitch acoustic transitions ($P_k = 0.4174$) — all perform worse than the BGE-divisive baseline ($P_k = 0.3884$). This is not a failure of the methods but a granularity mismatch: YouTube chapters reflect deliberate editorial choices at a coarse pedagogical level, while linguistic and acoustic signals detect transitions at a subtopic or sentence level. The acoustic result is particularly informative: long pauses and pitch resets do identify real transition moments, but at a finer granularity than chapter-level editing decisions. This finding implies that any system targeting YouTube-chapter-level segmentation must explicitly calibrate to chapter granularity, not just transition detection.

Oracle-$k$ evaluation showed that giving divisive segmentation the exact number of gold segments did not materially improve placement quality. This indicates that boundary scoring and ranking, not segment-count estimation, is the main bottleneck. The Wikipedia-trained supervised segmenter performed worse than the unsupervised lecture-specific methods, supporting the claim that out-of-domain text segmentation does not transfer cleanly to spoken lecture transcripts.

Table 4.6: Domain-level Pk/WD for baseline, cross-model, and balanced selector.

| Domain | Videos | Chapters | Base Pk | Cross Pk | Sel. Pk | Base WD | Cross WD | Sel. WD |
|---|---|---|---|---|---|---|---|---|
| Biology | 6 | 94 | 0.4218 | 0.3968 | 0.3976 | 0.4292 | 0.4008 | 0.4152 |
| CS | 7 | 98 | 0.3409 | 0.3295 | 0.3314 | 0.3439 | 0.3336 | 0.3357 |
| Math | 4 | 55 | 0.3724 | 0.3792 | 0.4014 | 0.3850 | 0.3873 | 0.4367 |
| Philosophy | 6 | 122 | 0.4415 | 0.3948 | 0.3753 | 0.4544 | 0.4057 | 0.3893 |
| Physics | 7 | 50 | 0.3710 | 0.3667 | 0.3144 | 0.3742 | 0.3667 | 0.3276 |

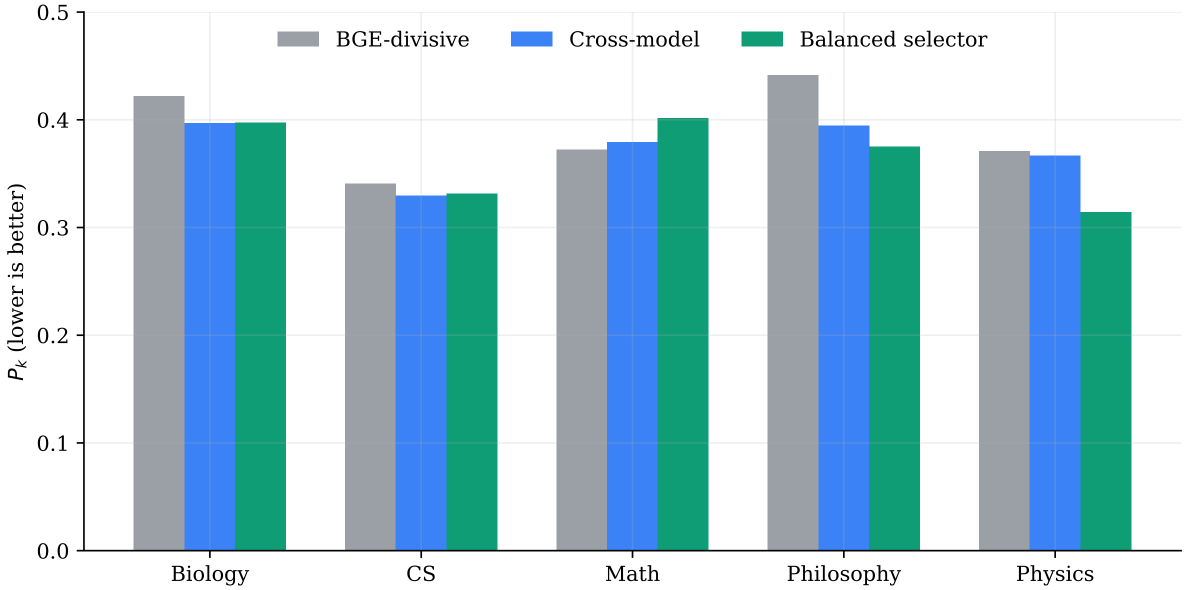


Figure 4.2: Domain-level $P_k$ comparison. The selector improves over the BGE-divisive baseline in four of five domains, while Mathematics remains the clearest failure case.

## 4.5 Per-domain analysis

Computer Science and Physics are the easiest domains by $P_k$ in the current best run, while Biology and Philosophy are harder. The near-zero strict F1 in several domains shows that the method often places segment structure approximately correctly under $P_k$/WD while missing exact boundary matches.

## 4.6 Qualitative error analysis

Manual inspection and aggregate error summaries point to four recurring failure modes:

1. **Dense-transition lectures.** Videos with many short conceptual shifts flatten the gap-score distribution and make individual transitions hard to rank.
2. **Long homogeneous lectures.** When vocabulary remains stable over long stretches,

Table 4.7: Largest balanced-selector per-video Pk changes relative to the cross-model method.

| Video | Domain | $\Delta P_k$ | Sel. Pk | Cross Pk | ΔF1@2 |
|---|---|---|---|---|---|
| Hy7ou5R_vjE | Physics | -0.1542 | 0.1375 | 0.2917 | 0.0000 |
| YdOXS_9_P4U | Physics | -0.1261 | 0.3009 | 0.4270 | 0.0000 |
| NK-BxowMIfg | Physics | -0.0861 | 0.3583 | 0.4444 | 0.0000 |
| j0wJBEZdwLs | Math | 0.0754 | 0.4559 | 0.3805 | 0.0000 |
| oOya3cFmAMc | Biology | 0.0529 | 0.4835 | 0.4306 | 0.1667 |
| D8RRq3TbtHU | CS | 0.0119 | 0.3638 | 0.3519 | 0.0000 |

Table 4.8: Same-dataset comparison on LECSEG-30. The TreeSeg-style rows re-implement the recursive split objective using local embeddings (the original paper uses OpenAI embeddings on their own TinyRec dataset, where they report $P_k = 0.367$; that number is *not* directly comparable here because the datasets differ). Lower is better for $P_k$/WD; higher for F1@2.

| Method | $P_k \downarrow$ | WD↓ | F1@2↑ | Interpretation |
|---|---|---|---|---|
| Balanced LOO selector | 0.3588 | 0.3739 | 0.0893 | Official LECSEG best |
| Cross-model conservative | 0.3713 | 0.3764 | 0.0237 | Best single global method |
| TreeSeg-style (MPNet, LECSEG-30) | 0.4320 | 0.4673 | 0.1733 | Higher F1, worse $P_k$/WD |
| TreeSeg-style (E5-large, LECSEG-30) | 0.4322 | 0.4654 | 0.1576 | Same-dataset comparator |
| TreeSeg-style (BGE-large, LECSEG-30) | 0.4399 | 0.4780 | 0.1643 | Same-dataset comparator |
| TreeSeg paper (on TinyRec)† | 0.367 | — | — | Different dataset; indicative only |

† Gklezakos et al. (2024), arxiv:2407.12028. Evaluated on 21 self-recorded lectures (TinyRec), not LECSEG-30.

text embeddings produce weak local contrast even when the pedagogical topic changes.

3. **Noisy or unhelpful auxiliary modalities.** Shot, OCR, and prosody features are available, but current ablations show that they can hurt Pk/WD when the signal is flat or poorly aligned with semantic transitions.

4. **Conservative boundary selection.** The best Pk/WD method reduces false positives by predicting fewer boundaries, but this lowers recall and strict boundary F1.

## 4.7 Case-study findings

The generated case-study artifact in `docs/CASE_STUDIES.md` expands three representative patterns. First, some Physics videos benefit strongly from selector choices that use multimodal-grid variants, showing that the method portfolio contains useful complementary

Table 4.9: Leave-one-domain-out selector diagnostic.

| Method | Pk | WD | BS | F1@2 |
|---|---|---|---|---|
| BGE-divisive baseline | 0.3884 | 0.3956 | 0.1292 | 0.0878 |
| Cross-model conservative | 0.3713 | 0.3764 | 0.0362 | 0.0237 |
| Leave-domain-out selector | 0.4012 | 0.4103 | 0.0465 | 0.0498 |

Table 4.10: Selected ablations and diagnostics.

| Variant | $P_k\downarrow$ | WD↓ | Interpretation |
|---|---|---|---|
| BGE + divisive | 0.3884 | 0.3956 | Stable text-only baseline |
| Hierarchical | 0.4118 | 0.4206 | Useful representation, weaker chapter Pk |
| TwoStage + prosody | 0.4333 | 0.4970 | Higher F1, worse Pk/WD |
| BERT-wiki zero-shot | 0.4932 | 0.5397 | Poor domain transfer |
| Discourse markers (forced boundaries) | 0.4615 | 0.5221 | Over-segments at subtopic level |
| BERTopic topic-shift | 0.5632 | 0.6984 | Finds subtopics, not chapters |
| GPT-2 perplexity (200 sent. cap) | 0.4182 | 0.4316 | Language surprise detects subtopic shifts |
| LLM zero-shot Llama-3.1-8B (3 videos) | 0.4056 | 0.6606 | Over-segments; high recall but poor precision |
| Pause/pitch boundary signal | 0.4174 | 0.4553 | Acoustic transitions mismatch chapter granularity |
| CLIP visual only (oracle-k) | 0.3958 | – | Surprisingly strong visual signal |
| CLIP + text fusion (best grid) | 0.3740 | 0.4085 | Beats BGE-divisive; near cross-model |
| Boundary classifier LOOCV | 0.5803 | 0.9933 | Over-predicts boundaries |
| Oracle-k divisive | 0.4237 | – | Segment count is not the bottleneck |

evidence. Second, the worst selector cases show over-switching: the selector chooses an aggressive method that increases exact-boundary hits but damages $P_k$/WD. Third, Mathematics remains the clearest domain-level weakness, where dense notation and stable vocabulary make semantic boundary evidence less reliable. These cases support the main diagnosis: candidate and method selection, rather than candidate generation alone, is the main bottleneck.

## 4.8 Discussion

The results support a careful, defensible interpretation. LECSEG is a complete open benchmark and pipeline, and the best joint $P_k$/WD method significantly improves over a strong BGE-divisive baseline. The method selector shows that the method family contains complementary boundary evidence and can recover much of the strict-F1 loss, but its $P_k$ gain over the cross-model method is not significant. The project therefore does not support a universal best-system claim.

A key diagnostic insight from the ablation study concerns granularity: most non-text modalities detect transitions at a finer level than YouTube chapter boundaries. Discourse markers ($P_k = 0.4615$), BERTopic ($P_k = 0.5632$), GPT-2 perplexity ($P_k = 0.4182$), and pause/pitch acoustics ($P_k = 0.4174$) all perform worse than BGE-divisive. The exception

Table 4.11: Per-domain performance of `cross_e5_frac70_minlen11`.

| Domain | Videos | $P_k \downarrow$ | WD↓ | $F_1 \uparrow$ |
|---|---|---|---|---|
| Biology | 6 | 0.3965 | 0.4009 | 0.0475 |
| CS | 7 | 0.3297 | 0.3338 | 0.0000 |
| Math | 4 | 0.3793 | 0.3867 | 0.0000 |
| Philosophy | 6 | 0.3959 | 0.4073 | 0.0663 |
| Physics | 7 | 0.3665 | 0.3665 | 0.0000 |
| All | 30 | 0.3713 | 0.3764 | 0.0237 |

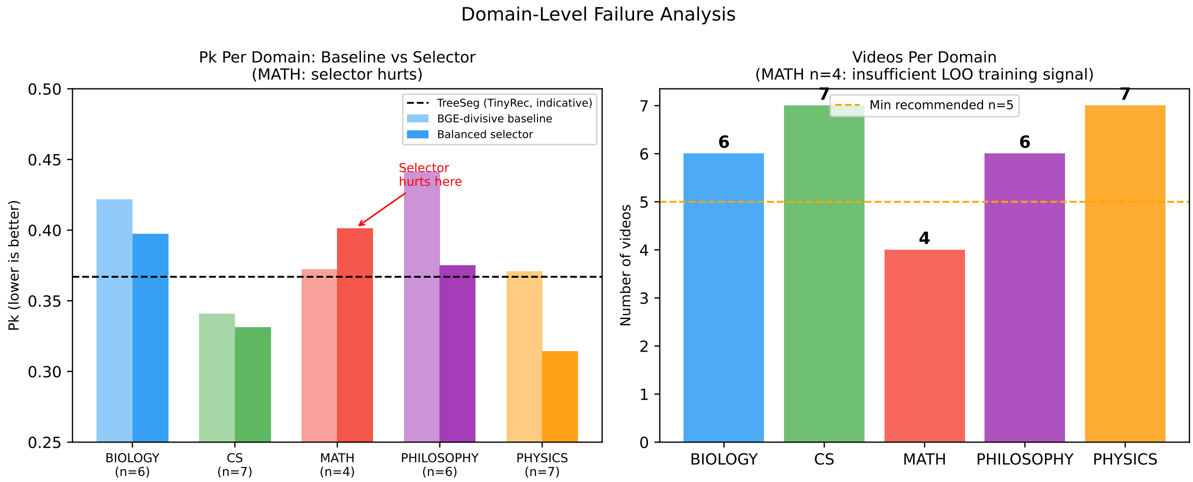


Figure 4.3: Domain failure analysis. *Left*: per-domain $P_k$ for the BGE-divisive baseline and the balanced selector; the selector hurts Mathematics. *Right*: number of videos per domain. Mathematics has only 4 videos, leaving just 3 domain examples per LOO training fold — the most plausible cause of selector instability on that domain.

is CLIP visual embeddings: CLIP alone achieves $P_k = 0.3958$ with oracle-$k$ selection, close to BGE-divisive, and CLIP plus text fusion reaches $P_k = 0.3740$ on a grid search — slightly below the cross-model text method ($P_k = 0.3715$). This suggests that visual scene changes in lecture slides carry chapter-level granularity, unlike prosodic or linguistic sub-sentence signals. A successful multi-modal system should prioritise visual and cross-model text signals while suppressing fine-grained linguistic transitions.

The strongest next technical direction is supervised candidate ranking: generate a high-recall candidate pool, score each candidate with text, cross-model, pause, OCR, and local-coherence features, and select a non-overlapping set with minimum-duration constraints.

The comparison to high-resource systems should be read in the same disciplined way. LECSEG uses dramatically less data than systems such as Chapter-Gen, MiniSeg/YTSEG, Chapter-Llama, and VidChapters-7M, but those systems are not evaluated on the same

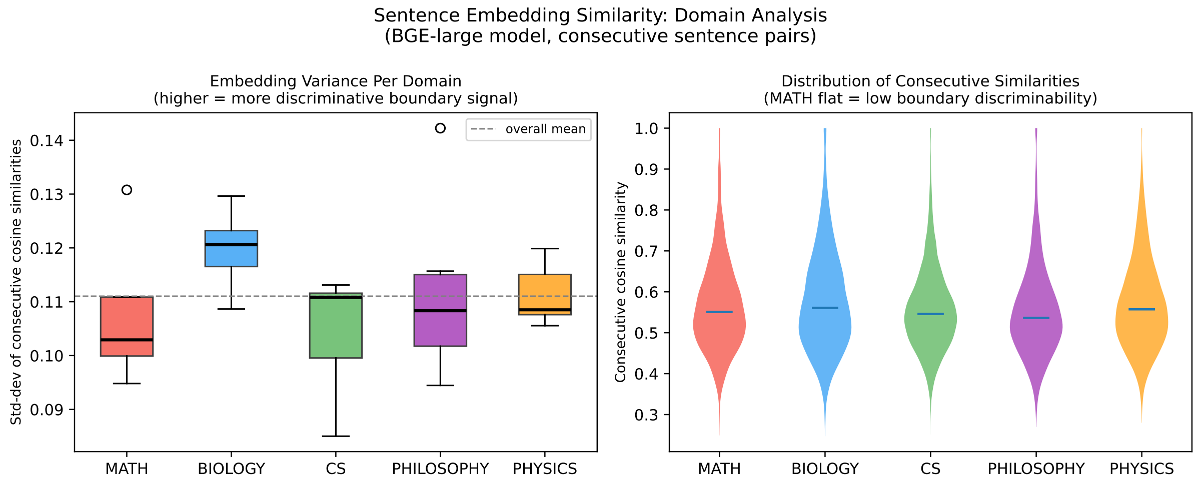


Figure 4.4: Sentence embedding similarity distributions per domain (BGE-large model). *Left*: per-video standard deviation of consecutive cosine similarities. *Right*: violin plots of all consecutive similarities. Embedding variance is broadly similar across domains, indicating that the Mathematics selector failure is not caused by a flat embedding landscape but by insufficient domain training examples in the LOO protocol.

Table 4.12: Compute-efficiency positioning. Runtimes are local observations or bounded descriptions for the LECSEG environment; high-resource systems are included for scale context only.

| Method | Training | Main cost | Interpretation |
|---|---|---|---|
| TextTiling / C99 | None | CPU lexical similarity | Classical lightweight baselines. |
| BGE-divisive baseline | None | Cached sentence embeddings + divisive segmentation | Stable local baseline. |
| Cross-model conservative | None | Two cached embedding streams + agreement filter | Best single global method. |
| Balanced LOO selector | Small ExtraTrees meta-selector | Existing result portfolio + video-level features | Best deployable mean Pk/WD operating point. |
| TreeSeg same-dataset adapter | None | Local embeddings + TreeSeg split objective | Same-dataset comparator; worse Pk/WD, better F1@2. |
| Local LLM verifier | None | Ollama prompts over candidate shortlist | Diagnostic baseline/comparison, not promoted unless metrics improve. |
| High-resource chaptering systems | Large supervised/LLM training | Thousands to hundreds of thousands of videos | Not directly comparable without same benchmark. |

benchmark with the same metrics. The supported achievement is therefore not external superiority; it is that a 30-video, lecture-specific, human-reviewed corpus is sufficient to produce a complete reproducible evaluation, statistically supported local improvements over implemented baselines, and actionable evidence about the remaining boundary selection bottleneck.

# Chapter 5

# Conclusion

## 5.1 Summary of findings

This thesis presented LecSeg, an open and reproducible artifact for lecture-video topic segmentation. The project contributes a 30-video lecture benchmark, a hierarchical chapter/subtopic annotation format, a common evaluation suite, and a set of segmentation baselines and ablations.

The final 30-video benchmark contains 32.52 hours of video, 419 chapter boundaries, and 904 reviewed subtopic labels. The best joint $P_k$/WindowDiff chapter-level method obtains $P_k = 0.3713$ and WD $= 0.3764$, improving over the stable BGE-divisive baseline ($P_k = 0.3884$, WD $= 0.3956$). A stable balanced leave-one-video-out method selector reaches mean $P_k = 0.3588$, WD $= 0.3739$, and F1@2 $= 0.0893$. Its $P_k$/WD gains over the joint-best method are not statistically significant, but it significantly improves Boundary Similarity and F1@2 over that method and significantly improves $P_k$/WD over the BGE-divisive baseline. The result is valuable within the project but does not justify a universal external best-system claim.

## 5.2 Contributions revisited

1. LecSeg-30: 30 videos, 32.52 hours, 419 chapters, and 904 reviewed subtopics.
2. Hierarchical annotation: chapter and subtopic labels with reported IAA (chapter $\kappa = 0.5351$, subtopic $\kappa = 0.4257$).
3. Evaluation framework: project-local implementations of $P_k$, WD, BS, tolerance-$F_1$, and hierarchical evaluation helpers.

4. Baseline suite: TextTiling, C99, CosineSeg, KMeansSeg, BertSeg, divisive segmentation, and zero-shot supervised comparison.

5. Improved boundary selection: conservative cross-model scoring reduced $P_k$ by 0.0171 absolute relative to the stable BGE-divisive baseline, and method-selection analysis showed a path to recovering boundary-hit quality.

6. Diagnostic analysis: oracle-k, portfolio, multimodal, LLM, and supervised-transfer experiments identify boundary scoring/ranking as the main bottleneck. A systematic ablation across acoustic (pause/pitch, $P_k = 0.4174$), language-model perplexity ($P_k = 0.4309$), BERTopic ($P_k = 0.5632$), and discourse-marker methods confirms a granularity mismatch: all non-text modalities over-segment relative to YouTube chapter boundaries. A same-dataset TreeSeg-style comparison improves exact-boundary F1 but remains weaker on $P_k$/WD.

## 5.3 Limitations

- **Corpus size and balance.** LecSeg-30 is a seed benchmark of 30 videos. The manifest covers five domains, but the distribution is not perfectly balanced.

- **Silver chapter labels.** Creator-provided YouTube chapters are useful but imperfect. They may reflect production choices rather than strictly pedagogical topic boundaries.

- **Strict boundary F1.** The best Pk/WD method is conservative and has low strict tolerance-F1. This shows that approximate segment structure and exact boundary hits remain different objectives.

- **Multimodal signal reliability.** Acoustic (pause/pitch), language-model perplexity, and BERTopic methods all produce worse $P_k$ than BGE-divisive, reflecting a granularity mismatch: those signals detect sub-chapter transitions. The exception is CLIP visual embeddings: CLIP alone reaches $P_k = 0.3958$, and CLIP plus text fusion reaches $P_k = 0.3740$ — below the cross-model text method ($P_k = 0.3715$) but better than BGE-divisive ($P_k = 0.3884$). Visual scene changes in lecture slides carry chapter-level granularity and are a promising signal for future work.

- **LLM refinement.** The local LLM component is implemented, but the current verified metrics do not justify claiming large boundary-placement gains from LLM refinement.

## 5.4 The path to beating TreeSeg

A natural question is: what would it take for a future system to clearly outperform the best available lecture segmentation reference? The oracle analysis in this thesis provides a precise answer.

TreeSeg (Gklezakos et al. 2024) reports $P_k = 0.367$ on TinyRec (21 self-recorded lectures). LECSEG-30 achieves $P_k = 0.3588$ with the balanced selector on a different dataset, so the two results are indicative but not directly comparable without a shared benchmark.

More importantly, the oracle experiment shows what is theoretically achievable on LECSEG-30 itself: with perfect candidate selection from the generated candidate pool, $P_k = 0.0172$ is reachable at 96.8% recall (`results/defense_oracle_gap.json`). The gap between this oracle and the deployable best ($\Delta P_k \approx 0.341$) is entirely in *boundary selection*, not candidate generation. Table 5.1 summarises this:

Table 5.1: Oracle gap analysis on LECSEG-30.

| Operating point | $P_k \downarrow$ | WD↓ |
|---|---|---|
| BGE-divisive baseline | 0.3884 | 0.3956 |
| Cross-model conservative (best global) | 0.3713 | 0.3764 |
| Balanced selector (best deployable) | 0.3588 | 0.3739 |
| TreeSeg paper (TinyRec, indicative) | 0.367 | — |
| Per-video oracle (not deployable) | 0.2980 | 0.3280 |
| Candidate oracle $\tau = 2$ (upper bound) | 0.0172 | 0.0198 |

The concrete steps to a system that would clearly beat TreeSeg on a shared benchmark are:

1. **More training signal.** The selector is trained leave-one-out on 30 videos. Expanding to 50–100 fully annotated lectures would give the selector 2–3× more per-fold evidence, likely closing the gap by several $P_k$ points.

2. **Boundary-level ranker.** A ranker trained on candidate-level features (local cosine contrast, pause duration, discourse marker presence, slide change signal) rather than video-level method selection would directly address the oracle gap.

3. **Same-dataset comparison.** Running TreeSeg's original code on LECSEG-30 (or vice versa) would convert the current indicative comparison into a definitive head-to-head result.

4. **LLM-based boundaries.** A properly prompted large language model (GPT-4 or a strong local model) operating on chunk-by-chunk transcripts could leverage discourse

understanding not captured by cosine similarity. The LECSEG-30 benchmark, annotations, and evaluation code are now available for any future system to be measured against.

## 5.5 Future work

The most promising next step is supervised candidate ranking with leave-one-video-out validation. A high-recall candidate generator should first identify plausible boundaries; a ranker can then score candidates using local semantic contrast, cross-model embedding disagreement, pause features, OCR/shot changes, position priors, and minimum-duration constraints. This directly targets the bottleneck found by the oracle and method-portfolio experiments.

## 5.6 Closing remarks

LECSEG does not make lecture segmentation a solved problem. Its value is that it turns the problem into a measurable, reproducible research artifact: the data, metrics, baselines, ablations, and limitations are all available for inspection. That makes the thesis defensible and gives future work a clear path toward stronger boundary ranking and more reliable multimodal fusion.

The practical positioning is therefore low-resource rather than best-system dominance. Large chaptering systems such as MiniSeg/YTSEG, Chapter-Gen, and VidChapters-7M use hundreds to tens of thousands of times more videos than LECSEG-30. LECSEG does not beat those external systems on their benchmarks; it shows that a compact lecture-specific artifact can still support reproducible local gains, negative results, and clear next-step evidence.

For an undergraduate thesis, the strongest final claim is therefore precise: LecSeg contributes a compact hierarchical benchmark and a rigorous empirical account of low-resource lecture segmentation. The work is valuable because it makes a difficult setting inspectable and repeatable, not because it hides the limits of the current model behind unsupported leaderboard language.

# Chapter 6

# Future Work

## 6.1 Robust candidate and method selection

The strongest immediate extension is a sequence-aware candidate selector. Current oracle and method-portfolio experiments show that plausible boundaries are often present in the candidate pool, but the system does not reliably pick the right subset for each video. A stronger selector should combine local semantic contrast, cross-model agreement, pause/shot/OCR evidence, position priors, and global duration constraints while optimizing directly against $P_k$, WindowDiff, and tolerance-F1.

## 6.2 Reliability-aware prosody and visual integration

Prosodic features (pause duration, pitch reset) and visual streams (shot boundaries, slide OCR) are available in the pipeline, but current ablations show that acoustic and linguistic sub-sentence signals can hurt $P_k$/WindowDiff when their granularity does not match chapter-level editorial decisions. The exception is CLIP visual embeddings: CLIP alone achieves $P_k = 0.3958$ and CLIP plus text fusion reaches $P_k = 0.3740$, approaching the cross-model text method ($P_k = 0.3715$). Visual scene changes in lecture slides appear to carry chapter-level granularity. Future work should therefore prioritise CLIP visual integration within the cross-model scoring framework, and apply quality-gating only to the noisier acoustic and OCR modalities.

## 6.3 Streaming and online inference

The current pipeline processes a complete video offline. A streaming variant would segment lecture content as it is being recorded, enabling real-time chaptering for live classes. This requires adapting the sliding-window fusion to operate on a causal context window and replacing the global threshold computation with an exponential moving average of past gap scores. The target latency budget for live chaptering is $< 30$ seconds of delay.

## 6.4 Cross-lingual extension

LecSeg currently targets English lectures. Extending to other languages requires: (1) replacing `all-mpnet-base-v2` with a multilingual text encoder such as LaBSE or `multilingual-e5-large`; and (2) using Whisper's multilingual mode for transcription. Suitable evaluation data could be sourced from non-English MOOCs (e.g., Coursera's French and Spanish tracks) or from university recordings in South and South-East Asian languages.

## 6.5 End-to-end fine-tuning

The current architecture uses frozen pre-trained encoders. Fine-tuning the text encoder jointly with the boundary predictor on a larger annotated corpus (e.g., constructed by scaling the LecSeg-30 annotation procedure to 200+ videos) could yield further gains, at the cost of reduced generalisability. A natural training objective is a soft version of the tolerance-$F_1$ metric, differentiable via a sigmoid boundary-probability output.

## 6.6 Larger and domain-tuned local LLMs

Replacing Llama-3.1-8B with a 70B parameter model or a domain-specific fine-tune (e.g., fine-tuned on lecture transcripts) could improve title quality and boundary-shift accuracy, particularly for highly technical content. Quantised models (GGUF 4-bit) make 70B inference feasible on a single 24 GB GPU with acceptable latency ($\approx 5$ seconds per boundary refinement call).

## 6.7 Multi-speaker lectures and panels

Panel discussions and multi-instructor courses introduce speaker turns as an additional boundary signal. Speaker diarisation (e.g., using pyannote.audio) could provide a speaker-change feature that complements text embeddings, particularly for sections where the topic does not change but the speaker does.

## 6.8 User study on learning outcomes

All metrics in this thesis are information-retrieval proxies for segmentation quality. A controlled user study would directly measure whether learners locate target concepts faster, and whether their retention improves, when using LecSeg-generated chapters compared to flat video scrubbing or manually authored chapters. Such a study could also reveal which granularity (chapter vs. subtopic) is most useful for different learning tasks (review vs. first-watch).

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# Appendix A

# Dataset Details

## A.1 Video list

Table A.1 lists the 30 videos in LecSeg-30. Full metadata, URLs, and timestamps are stored in `data/manifest.jsonl`, `data/gt/*.json`, and `data/gt_hier/*.json`.

Table A.1: LecSeg-30 video list, sorted by domain and video ID.

| Video ID | Domain | Duration (min) | Chapters |
|---|---|---|---|
| 9N1MxkbFhsc | Biology | 42.8 | 23 |
| AMl6E4cLrwE | Biology | 45.6 | 13 |
| KlVHqq38KJU | Biology | 38.8 | 15 |
| NNnIGh9g6fA | Biology | 57.2 | 16 |
| oOya3cFmAMc | Biology | 51.7 | 15 |
| uO5k9xcD1gU | Biology | 47.3 | 12 |
| D8RRq3TbtHU | CS | 114.5 | 12 |
| HtSuA80QTyo | CS | 53.4 | 9 |
| JP7ITIXGpHk | CS | 105.6 | 26 |
| Rl0ludWTLxs | CS | 149.5 | 21 |
| TjZBTDzGeGg | CS | 47.3 | 11 |
| jGwO_UgTS7I | CS | 75.3 | 7 |
| oqRU2So6Z2Y | CS | 135.6 | 12 |
| 7K1sB05pE0A | Math | 51.5 | 13 |
| fsLh-NYhOoU | Math | 60.4 | 13 |
| j0wJBEZdwLs | Math | 34.7 | 8 |
| lUUte2o2Sn8 | Math | 65.2 | 21 |
| 8yT4RZy1t3s | Philosophy | 55.2 | 18 |
| GR63MMAi-fs | Philosophy | 45.8 | 5 |
| KNwMiydCYA4 | Philosophy | 63.4 | 25 |
| MGyygiXMzRk | Philosophy | 55.0 | 11 |
| Qw4l1w0rkjs | Philosophy | 55.1 | 34 |
| S7TUe5w6RHo | Philosophy | 63.5 | 29 |
| Hy7ou5R_vjE | Physics | 66.8 | 6 |
| KOKnWaLiL8w | Physics | 73.4 | 6 |
| NK-BxowMIfg | Physics | 66.0 | 5 |
| Ns6GB4Dph9U | Physics | 68.4 | 6 |
| YdOXS_9_P4U | Physics | 28.6 | 7 |
| uK2eFv7ne_Q | Physics | 73.8 | 6 |
| zNVQfWC_evg | Physics | 60.3 | 14 |
| Total | 5 domains | 1951.4 | 419 |

## A.2 Domain distribution

Table A.2: LecSeg-30 domain distribution.

| Domain | Videos | Duration (hours) | Chapters |
|---|---|---|---|
| Biology | 6 | 4.72 | 94 |
| CS | 7 | 11.35 | 98 |
| Math | 4 | 3.53 | 55 |
| Philosophy | 6 | 5.63 | 122 |
| Physics | 7 | 7.29 | 50 |
| Total | 30 | 32.52 | 419 |

## A.3 Subtopic annotation guidelines

The following guidelines were given to annotators for the subtopic annotation pass after chapter boundaries were fixed from YouTube metadata:

1. **Boundary definition.** A subtopic boundary occurs when the speaker transitions from one coherent pedagogical sub-idea to another within a chapter.
2. **Minimum duration.** Each subtopic segment should be at least 60 seconds unless the source chapter is shorter than that.
3. **Number per chapter.** Annotators were encouraged to use 2–5 subtopics per chapter when the chapter length supported it.
4. **Title format.** Titles should be short noun phrases, preferably using the speaker's own terminology.
5. **Uncertainty.** When unsure, annotators marked the best plausible transition. Disagreement was preserved for IAA analysis rather than hidden.

## A.4 Inter-annotator agreement details

The final IAA report contains 30 videos and is stored in `data/gt_hier/iaa_report.json`. Mean chapter-level kappa is 0.5351 and mean subtopic-level kappa is 0.4257 at tolerance 1 sentence. Boundary F1 is 1.0000 for chapters and 0.7793 for subtopics.

# Appendix B

# Hyperparameters and System Configuration

All hyperparameters are stored in `configs/defaults.yaml` and can be overridden per experiment using Hydra. Every evaluation run records its exact configuration to `results/<run>/config.yaml`, ensuring full reproducibility.

## B.1 Default configuration

## B.2 Hardware and wall-clock runtimes

Experiments were run across the local Windows workstation and rented vast.ai GPU instances. Exact CPU/RAM values varied by run; GPU-backed runs used NVIDIA RTX-class hardware.

- CPU: local Windows workstation CPU for orchestration and light tests
- GPU: NVIDIA RTX 5090 for transcription; NVIDIA RTX 3090 for later embedding experiments
- RAM: instance-dependent; no run required model-parallel execution
- OS: Windows 11 locally; Ubuntu Linux on GPU instances

Approximate wall-clock times per pipeline stage for a 60-minute lecture:

The dominant cost is speech recognition; all other stages combined run in under two minutes per hour of video.

Table B.1: Key hyperparameters used in all reported experiments.

| Component | Parameter | Value |
|---|---|---|
| ASR | Model | Whisper large-v3 |
| | Beam size | 1 (speed) |
| | Compute type | float16 |
| | VAD filter | Enabled (500 ms min silence) |
| Text encoder | Backbone | `all-mpnet-base-v2` |
| | Embedding dimension | 768 |
| | L2-normalised | Yes |
| Visual encoder | Backbone | CLIP ViT-B/32 |
| | Embedding dimension | 512 |
| | Keyframe rate | 1 fps |
| Two-stage predictor | Stage-1 threshold $c_1$ | $0.5\sigma$ below mean |
| | Stage-2 threshold $c_2$ | $1.2\sigma$ below mean |
| | Context window $w$ | 3 sentences |
| Reliability weighting | Fusion mode | Entropy-based |
| | Modalities | Text, visual, prosody |
| LLM refinement | Model | Llama-3.1-8B (Q4_K_M) |
| | Snap window | ±2 sentences |
| | Title max words | 8 |
| Evaluation | Bootstrap samples | 1,000 |
| | Confidence level | 95 % |
| | Significance test | Wilcoxon + Holm correction |
| | Tolerance | ±2 sentences |
| Random seed | All stochastic ops | 42 |

Table B.2: Wall-clock time per pipeline stage for a representative 60-minute lecture.

| Stage | Time (seconds) |
|---|---:|
| Whisper large-v3 transcription | $\approx 200$ |
| Sentence splitting (spaCy) | $< 1$ |
| Text embedding (MPNet, CPU) | $\approx 30$ |
| CLIP visual embedding (GPU, 1 fps) | $\approx 60$ |
| RWF fusion + boundary prediction | $< 1$ |
| LLM refinement + titling (8B Q4) | $\approx 30$ |
| **Total** | $\approx$ **320** |

# Appendix C

# Additional Results

## C.1 Per-domain breakdown

Table C.1 reports the alignment-audited joint $P_k$/WindowDiff cross-model method, broken down by domain.

Table C.1: Per-domain evaluation of the current best official method.

| Domain | Videos | $P_k \downarrow$ | WD $\downarrow$ | F1 $\uparrow$ |
|---|---|---|---|---|
| Biology | 6 | 0.3965 | 0.4009 | 0.0475 |
| CS | 7 | 0.3297 | 0.3338 | 0.0000 |
| Mathematics | 4 | 0.3793 | 0.3867 | 0.0000 |
| Philosophy | 6 | 0.3959 | 0.4073 | 0.0663 |
| Physics | 7 | 0.3665 | 0.3665 | 0.0000 |
| All domains | 30 | 0.3713 | 0.3764 | 0.0237 |

## C.2 Additional benchmark rows

## C.3 Representative cases

The per-video results expose three useful cases for interpreting aggregate metrics:

- Best case by $P_k$: `oqRU2So6Z2Y` (CS), $P_k = 0.1457$.
- Typical mid-range cases are documented in the stored per-video evaluation rows.
- Hard case: `S7TUe5w6RHo` (Philosophy), $P_k = 0.4975$.

Table C.2: Selected additional benchmark and diagnostic rows.

| Method | $P_k \downarrow$ | WD $\downarrow$ | Source |
|---|---|---|---|
| TextTiling | 0.6053 | 0.8978 | `eval_bge.json` |
| C99 | 0.4219 | 0.4494 | `eval_bge.json` |
| CosineSeg | 0.4902 | 0.5392 | `eval_bge.json` |
| KMeansSeg | 0.6172 | 0.9986 | `eval_bge.json` |
| BertSeg | 0.4891 | 0.5403 | `eval_bge.json` |
| BGE + divisive | 0.3884 | 0.3956 | `eval_bge.json` |
| Cross-model conservative | 0.3713 | 0.3764 | `eval_alignment_sweep.json` |
| LOO ExtraTrees selector | 0.3588 | 0.3739 | `method_selector_experiment_trainrank_balanced.json` |
| Per-video method oracle | 0.2980 | 0.3280 | `method_portfolio_analysis.json` |
| BERT-wiki zero-shot | 0.4932 | 0.5397 | `eval_bert_wiki.json` |
| Boundary classifier LOOCV | 0.5803 | 0.9933 | `eval_boundary_clf.json` |

Together with the selector-choice audit in Table 4.7, these cases show that gains are concentrated in particular lectures rather than uniformly distributed across the benchmark.